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# Gas diffusion layer regulated phosphoric acid distribution for durable high-temperature polymer electrolyte membrane fuel cells under high-current loading

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## Abstract

High-temperature polymer electrolyte membrane fuel cells (HT-PEMFCs) offer promising power sources for heavy-duty vehicles (HDV) due to their effective management of heat and water. However, the long-term stability of HT-PEMFCs is critically constrained by phosphoric acid (PA) electrolyte loss especially at high current working circumstances. Herein this work, gas diffusion layers (GDLs) of distinct porous structures were designed to regulate PA distribution under rigorous loading circumstance. Combined with high-current-density (HCD) accelerated stress tests (AST) at  $1.6 \text{ A cm}^{-2}$  and in-operando 3D X-ray microscopic (XRM) characterization to systematically elucidate the influence of GDL structure on acid-loss pathways and performance degradation, the GDL featuring a dense microporous layer (MPL) with low mesoporosity exhibited the superior anti-leaching capability, showing the lowest acid loss rate ( $2.1 \text{ ng cm}^{-2} \text{ h}^{-1} \text{ L}^{-1}$ ) and the smallest voltage decay ( $0.23 \text{ mV cycle}^{-1}$ ) during HCD AST. Mechanistic analysis reveals that the dense MPL effectively mitigates liquid PA migration through catalyst layer (CL) cracks, while the high porosity compensates the performance loss led by mass transport resistance. This work suggests that engineering the overall GDL structure is an effective route to achieving long-term stable HT-PEMFC operation at HCD, providing key theoretical guidance for the design of next-generation, high-performance and durable membrane electrode assemblies (MEAs) for the heavy-duty applications.

**Keywords** Gas diffusion layer, Heavy-duty power sources, High-temperature polymer electrolyte membrane fuel cells, Phosphoric acid, 3D X-ray microscopic

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## 1 Introduction

Clean hydrogen is positioned as the critical solution to break the "hard-to-abate" bottleneck in China's carbon neutrality pathway [1]. However, realizing its large-scale deployment hinges on the development of efficient and reliable hydrogen conversion technologies. As an electrochemical energy conversion technology, HT-PEMFCs directly generate electricity and high-quality heat from hydrogen at temperatures above 160 °C, which confers inherent benefits including robust CO tolerance and simplified system management due to the absence of liquid water. These characteristics make HT-PEMFCs exhibit significant potential in fields such as heavy-duty trucks [2], aviation [3], and combined heat and power generation [4]. To meet the stringent requirements for the compactness and economy of power systems in commercial applications, continuously increasing the specific power densities and long-term durability of HT-PEMFCs has become the core focus of current research [5]. One key approach to achieving this goal is to enable the fuel cell to operate stably at higher current densities (HCD) to obtain greater output power from a single cell.

However, operating at HCD significantly accelerates the loss of PA electrolyte, severely impacting the durability and reliability of the fuel cells. Currently, the mechanism of PA loss can be divided into three main categories: loss during the MEA hot-press process, gaseous phase loss of cathode PA, and liquid phase migration of anode PA. Zucconi et al. [6] used in-situ X-ray imaging to discover that excessive hot-press pressure during assembly can force PA migrating into the GDL. Regarding gaseous phase loss, research teams led by Schmidt [7] and Li [8] systematically studied the effects of current density, gas flow rate, and temperature on PA loss behavior. Their findings indicated that both increased gas flow rate and temperature accelerate PA evaporation, while the water produced by increased current density washes away free PA from the cathode side. Kim's team [9] further explored the impact of stoichiometric ratio and inlet humidity and, through theoretical calculations, proposed a water replacement mechanism [10] to explain the exponential pattern of cathode gaseous phase loss rate, which initially increases rapidly and then slows down. In terms of anode liquid phase electromigration [11], Zeis' team studied the mode of PA intrusion into the GDL [12] and investigated the impact of pore size distribution and MPL morphology on PA intrusion [13]. They pointed out that as the current density increases,  $\text{H}_2\text{PO}_4^-$  migrate toward the anode under the influence of the electric field. Subsequent imaging studies [14] further revealed that PA preferentially forms loss pathways through overlapping cracks between the MPL and CL. The width, continuity, and accessibility of these pathways critically influence

the loss rate. Under high current density, PA loss in HT-PEMFCs is markedly accelerated via both cathodic gaseous loss and anodic liquid-phase electromigration. Since anodic migration is more importantly related to electrolyte depletion and lifetime decay, targeted optimization of the anode GDL is crucial for improving long-term stability.

To address the issue of PA loss, existing strategies have diversified. On the material level, the ion-pair membrane developed by Kim's team [15] enhances the anchoring of PA dihydrogen ions by introducing positively charged functional groups, effectively suppressing acid loss. Subsequently, polyaromatic piperidine systems with stable cationic groups were extended from the AEMFC field [16, 17] to HT-PEMFC, achieving fixation of PA dihydrogen ions. In contrast, researchers [18] used electrospun nanolayers to block anode PA loss, but this method sacrifices mass transport performance [19] and struggles to form an effective triple-phase interface. Other studies [20], drew on experiences from low-temperature proton-exchange-membrane fuel cells (LT-PEMFCs) and helpfully investigated the impact of anode MPL structure on PA leaching behavior. Although various solutions have been proposed in existing research to address PA loss, systematic studies under HCD conditions ( $\geq 1 \text{ A/cm}^2$ ) are still insufficient. Under these conditions, the rapid degradation of fuel cell performance and lifespan has become a major technical bottleneck. Therefore, developing a GDL structure that can ensure the stable operation of HT-PEMFCs has become an urgent need for engineering applications.

Recent studies on the stability of HT-PEMFCs indicate that under low current density conditions ( $\leq 0.2 \text{ A cm}^{-2}$ ), optimized PA-doped PBI-based membrane electrode assemblies can achieve extremely low voltage decay rates ( $0.5\text{--}2 \mu\text{V h}^{-1}$ ) [21], with the longest steady-state operational lifetimes exceeding 13,000–18,000 h [22–24]; at medium current densities ( $0.2\text{--}0.5 \text{ A cm}^{-2}$ ), the decay rate dramatically increases to  $2\text{--}25 \mu\text{V h}^{-1}$  [25], with typical lifetimes reduced to 1000–5000 h [26, 27]; when the current density exceeds  $0.5 \text{ A cm}^{-2}$  (especially  $>0.8 \text{ A cm}^{-2}$ ), accelerated PA loss and flooding effects become pronounced, causing the decay rate to sharply rise above  $25\text{--}100 \mu\text{V h}^{-1}$  [27], with actual durability lifetimes typically below 1000 h [25] and irreversible performance losses occurring within mere hundreds of hours. For every  $0.3\text{--}0.4 \text{ A cm}^{-2}$  increase in current density, the PA loss rate can rise by 2–10 times, posing the primary bottleneck [28] for achieving high power density and long-lifetime operation in HT-PEMFCs, especially for heavy-duty applications, such as UAVs and trucks, requiring ultra-high mass/volume power density. Compared with previous studies focusing primarily on

qualitative descriptions of electrolyte redistribution, the objective of this work is not to develop new GDL materials, but to systematically investigate how existing GDL microstructures regulate PA redistribution under ultra HCD conditions.

In this work, based on the principle of capillary pressure regulation of PA migration toward the anode, three GDLs with different structures were designed and characterized. The optimized performance and stability of the HT-PEMFCs with different GDLs at anodes under dynamic operating conditions was investigated, and the PA loss was simultaneously analyzed to establish a correlation between structure and fuel cell stability. Via an in-operando XRM characterization device developed in-house, dynamic observations of the PA distribution within a single cell were conducted under operating conditions, providing experimental evidences that electromigration plays an important role in driving PA redistribution under ultra HCD conditions in HT-PEMFCs.

## 2 Experimental

### 2.1 Reagents and materials

Three types of GDLs with different porous structures were purchased from Freudenberg (H23C8, H14C10, H23C14, labeled as DL-GDL, DH-GDL, LL-GDL, respectively). Composition for catalyst slurry Pt/C (60 wt% Pt, Johnson Matthey Chemicals), 60 wt% PtCo/C (47.1 wt% Pt, 14.5 wt% Co, Tanaka Kikinzoku Kogyo), PTFE dispersion (60 wt%, Shanghai 3F New Materials Technology), water and ethanol and PA-PBI membrane (Fumapem AMcl).

Three types of GDLs with distinct porous structures (Freudenberg, H23C8, H14C10, H23C14) were employed to investigate the influence of structure; they are hereafter denoted as DL-GDL, DH-GDL, and LL-GDL, respectively. Key materials for the catalyst layers included commercial Pt/C catalyst (Johnson Matthey Chemicals), PtCo/C alloy catalyst (Tanaka Kikinzoku Kogyo), and a PTFE dispersion (Shanghai 3F New Materials Technology) serving as a hydrophobic binder. Deionized water and ethanol were used as mixed solvents for all ink formulations. The membrane was a commercially available PA-doped polybenzimidazole membrane (Fumapem).

### 2.2 MEA fabrication

Membrane electrode assemblies were constructed by preparing gas diffusion electrodes followed by a hot-pressing process. The electrode fabrication procedure was as follows: First, the catalyst ink was formulated by mixing the respective catalyst (Pt/C for the anode, PtCo/C for the cathode), PTFE dispersion, water, and ethanol in pre-determined ratios. The mixture was ultrasonicated for 1 h to achieve a homogeneous dispersion. Subsequently,

the ink was uniformly deposited onto the corresponding GDL substrate via a spray-coating technique. To cure the PTFE binder and establish a stable porous structure, the coated electrodes were sintered at 300 °C for 30 min under a N<sub>2</sub> atmosphere. By precisely controlling the spray coating process, the final platinum loadings were fixed at 0.7 mg cm<sup>-2</sup> for the anode and 1.5 mg cm<sup>-2</sup> for the cathode, while the PTFE content in the catalyst layers was maintained at 20 wt% and 10 wt%, respectively.

For MEA assembly, a conventional three-layer hot-pressing method was adopted. The as-prepared GDEs and the PA-PBI membrane were aligned in a stack and then hot-pressed at 130 °C for 5 min to form an integrated unit. The resulting MEA possessed an active area of 20.25 cm<sup>2</sup> (4.5 cm × 4.5 cm).

## 3 Results and discussion

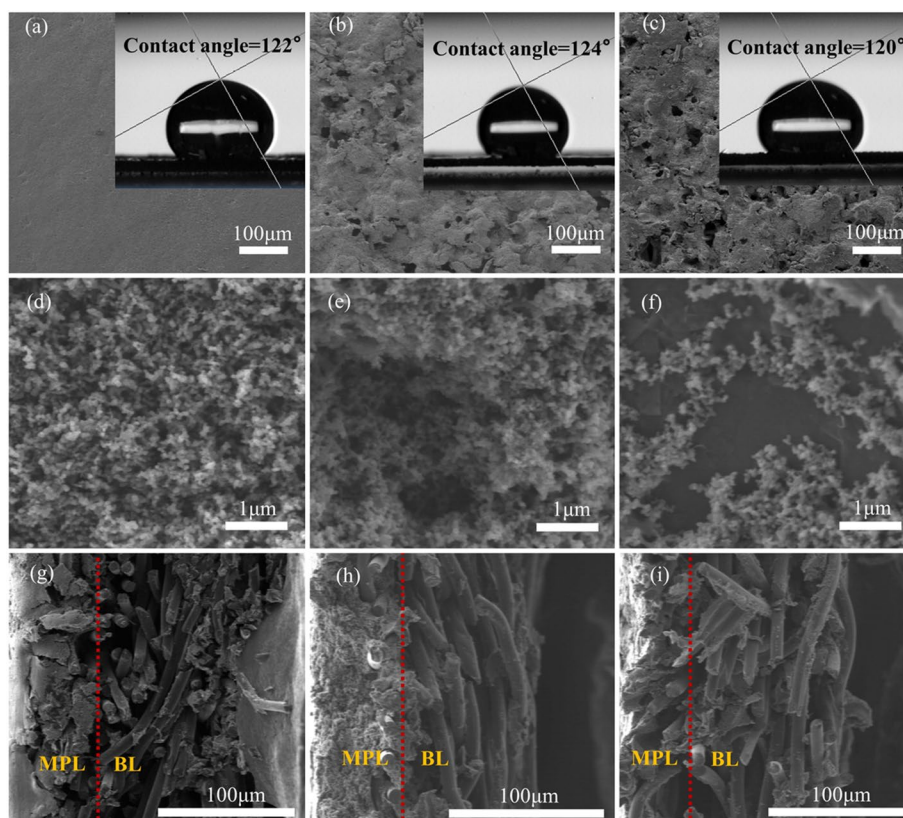
### 3.1 Structural and morphological analyses of GDLs

GDL, positioned between the CL and the bipolar plate, provides mechanical support, protects the membrane, and simultaneously mediates electron transport and uniform gas distribution, which comprises a backing layer (BL) and a MPL [29]. In HT-PEMFCs, GDL is additionally required to act as a regulator to help manage the distribution of PA. The resistance to acid intrusion is governed primarily by the capillary pressure balance between the pore network of the GDL and the PA. Drawing on strategies developed for LT-PEMFCs water management [30], cracked MPLs have been exploited to mitigate water flooding. Conversely, in HT-PEMFCs, where the operating temperature is higher and flooding is not a concern, the main challenge is PA electrolyte loss due to electromigration especially under high current loadings. Therefore, MPLs without cracks show better potential for retaining PA.

Hence, three crack-free GDLs with distinct MPL structures: i) dense MPL with low mesoporosity (DL-GDL); ii) dense MPL with high mesoporosity (DH-GDL); iii) loose MPL with low mesoporosity (LL-GDL) were selected to evaluate their impact on cell durability, as shown in Fig. 1. Because the interfacial tension  $\sigma$  between PA and the GDL material is an intrinsic property of the solid surface, it can be regarded as constant [31]; consequently, the capillary pressure is calculated according to Eq. (1) [32].

$$P_c = \frac{-4\sigma\cos\theta}{r} \quad (1)$$

A larger  $\theta$  (stronger PA repellency) and a smaller  $r$  together maximize the capillary barrier. As illustrated in Fig. 1a-c, after hydrophobization the PA contact angles of the three GDLs are almost identical.



**Fig. 1** Morphology of three GDLs. Surface SEM images (inset images: PA wetting behavior): **a, d** DL-GDL; **b, e** DH-GDL; **c, f** LL-GDL. Cross-sectional SEM images of different GDLs: **g** DL-GDL, **h** DH-GDL, **i** LL-GDL

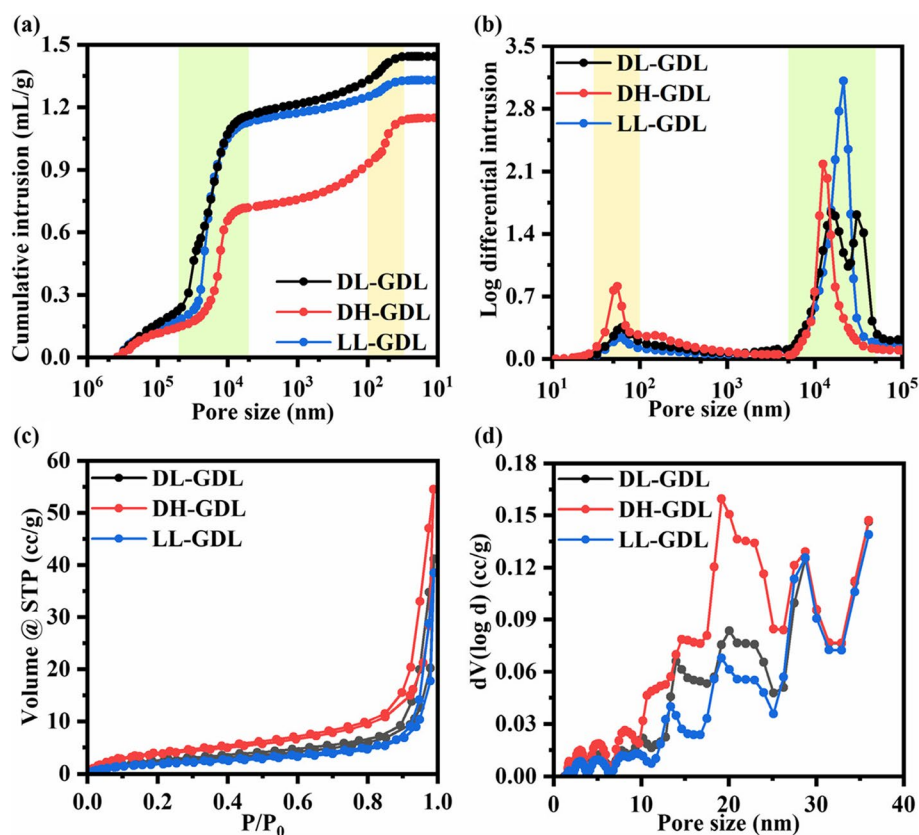
The surface morphology of the three GDLs, observed by field emission scanning electron microscopy (FE-SEM, Fig. 1), reveals distinct microstructural differences. The MPLs of the three GDLs differ markedly: DL-GDL appears almost pore-free, whereas pores are visible only at high magnification scale. At the same scale, DH-GDL and LL-GDL, own larger pores, showing less amorphous features. Cross-sectional SEM indicates MPL thicknesses of  $\sim 55 \mu\text{m}$  for all samples, with BL thicknesses of ca.  $170 \mu\text{m}$  (DL-GDL),  $105 \mu\text{m}$  (DH-GDL) and  $155 \mu\text{m}$  (LL-GDL), differences that influence pore-size distribution.

To obtain finer pore-structure information, mercury-intrusion porosimetry (MIP) and nitrogen adsorption-desorption isotherm Brunauer-Emmet-Teller (BET) method were conducted, as shown in Fig. 2. DH-GDL shows a cumulative intrusion of only  $0.5737 \text{ mL g}^{-1}$  between  $51,730$  and  $5126 \text{ nm}$ , versus  $0.9384 \text{ mL g}^{-1}$  for DL-GDL and  $0.9529 \text{ mL g}^{-1}$  for LL-GDL, as shown in Fig. 2a. This deficit reflects the  $\sim 50 \mu\text{m}$  thinner BL of DH-GDL. This also explains why DL-GDL appears to have a denser MPL in Fig. 1, whereas DH-GDL exhibits a higher overall mesopore fraction. Consistently, DH-GDL shows the smallest average pore diameter, median

pore diameter, and porosity, as summarized in Table S1. The corresponding peak at  $\sim 50 \text{ nm}$  in Fig. 2b corroborates this thinning and yields a higher gas permeability ( $52 \text{ Gurley s}$ , Table S1) relative to DH-GDL ( $92 \text{ Gurley s}$ , Table S3). Figure 2b and d further reveal that DL-GDL possesses a greater proportion of pores  $< 100 \text{ nm}$  than LL-GDL, and they have similar PA contact angle (Table S2), indicating a denser MPL and hence potential for suppressing liquid PA loss under HCD operation.

### 3.2 HT-PEMFC Performance at Beginning of Lifetime (BOL)

To elucidate the influence of GDL structure on cell performance and stability, three GDLs were evaluated by recording polarization and power-density curves under  $\text{H}_2/\text{air}$  and  $\text{H}_2/\text{O}_2$  feeds, as shown in Fig. 3. Under  $\text{H}_2/\text{air}$ , peak power densities (PPD) were  $343$ ,  $383$  and  $448 \text{ mW cm}^{-2}$  for DL-, DH- and LL-GDL equipped MEAs, respectively. With  $\text{H}_2/\text{O}_2$  condition, the PPDs are  $917$ ,  $1019$  and  $1153 \text{ mW cm}^{-2}$ , respectively. These differences correlate positively with GDL gas permeability: higher permeability affords higher voltage at high-current conditions dominated by mass transport. Switching to pure  $\text{O}_2$  alleviates cathodic mass-transfer polarization and narrows



**Fig. 2** Multiscale pore structure evaluation of the GDLs by complementary MIP and BET techniques. **a** MIP cumulative intrusion volume. **b** MIP-derived pore size distribution. **c** BET  $N_2$  physisorption isotherm at 77 K. **d** BET-based micropore size distribution

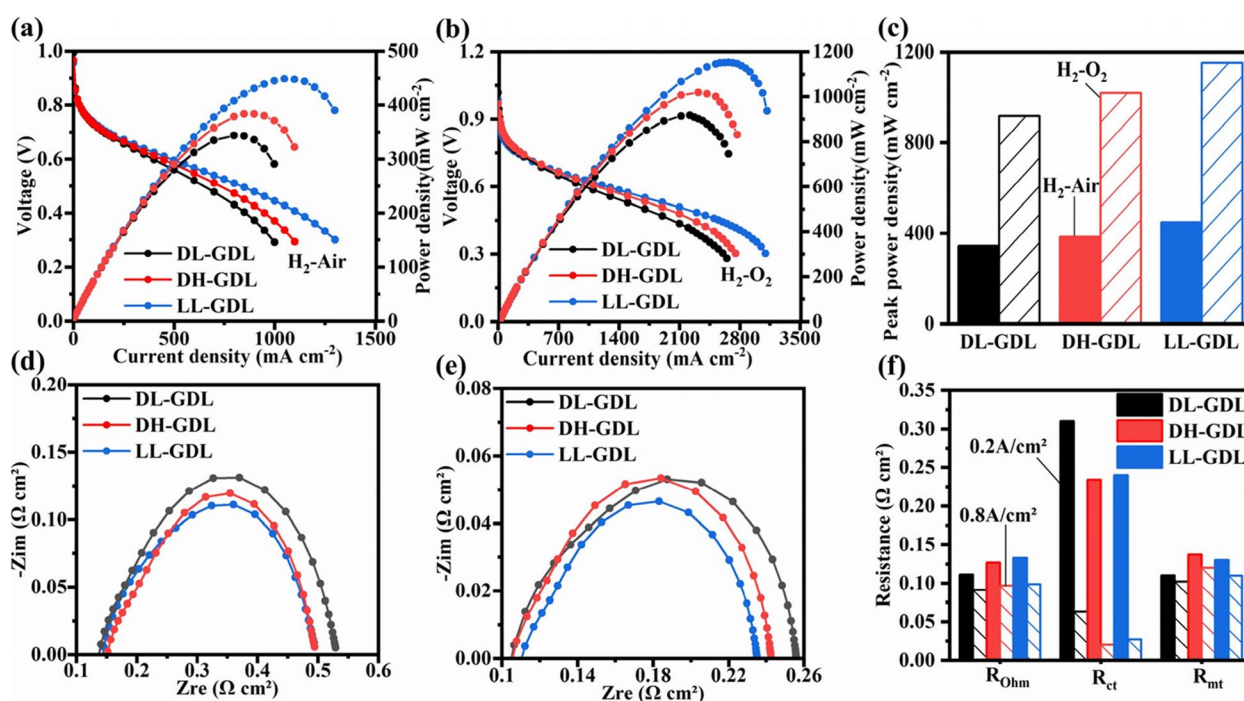
the performance gap, indicating that anodic transport resistance, though minor, remains operative. Electrochemical impedance spectra (EIS) (Fig. 3d–f) show that denser GDLs slightly reduce ohmic resistance but markedly increase charge-transfer resistance; cathodic mass-transfer limitations dominate. At  $0.8 \text{ A cm}^{-2}$  under excess  $H_2/O_2$ , mass transfer impedance primarily arises from PA-film local mass transfer resistance within the electrode, resulting in similar values for all three cells.

Although DL- and DH-GDL exhibit permeabilities of the same order (92 vs. 52 Gurley s) and their PPDs differ by  $\sim 10\%$  under both  $H_2/\text{air}$  and  $H_2/O_2$ , this similarity results from structural compensation: identical MPL thickness but a  $\sim 60 \mu\text{m}$  difference in BL thickness, along with distinct MPL pore-size distributions (Fig. 1a–b, g–h). Since Knudsen diffusion in the MPL is orders of magnitude slower than molecular diffusion in the BL, variations in BL thickness have little effect on overall permeability. LL-GDL, sharing MPL and BL thickness with DL-GDL, achieves markedly higher gas permeability (0.5 Gurley s) due to its unique MPL morphology, delivering the highest performance. Therefore, GDL optimization for enhanced performance requires tailoring MPL

morphology, pore-size distribution, and thickness, rather than simply reducing BL thickness. Even the anode, with modest mass-transfer demands, requires sufficient permeability to ensure rapid, uniform  $H_2$  distribution [33] and prevent the hydrogen-oxidation reaction from becoming a performance bottleneck.

### 3.3 AST test and HT-PEMFC performance at End of Lifetime(EOL)

To quantify anodic PA leaching under HCD conditions and accelerate the testing timeline [34], an ultra HCD AST was designed, based on correlational research and heavy-duty fuel cell durability protocols from the DOE [35]. Under  $H_2/O_2$  feed, cells were cycled between  $0.6 \text{ A cm}^{-2}$  and  $1.6 \text{ A cm}^{-2}$  (Fig. S8). The use of pure  $O_2$  alleviates cathodic mass-transfer resistance and enables stable operation at ultra HCD, allowing the influence of liquid PA migration to be isolated. The HCD segment accelerates PA migration, while the low current density (LCD) interval modulates migration velocity and promotes liquid expulsion through the GDL. This protocol therefore serves as an accelerated approach to evaluate electrolyte redistribution and durability under high-load operation



**Fig. 3** Initial performance metrics encompassing polarization behavior, power output, and impedance characteristics. **a, b** Voltage-current-power relationships for  $H_2/Air$  and  $H_2/O_2$  operation. **c** Maximum power density comparison. **d, e** Impedance spectra at low ( $200\ mA\ cm^{-2}$ ) and high ( $800\ mA\ cm^{-2}$ ) current densities under  $H_2/O_2$  at  $160\ ^\circ C$ . **f** Quantification of the corresponding ohmic, charge-transfer, and mass transport resistances

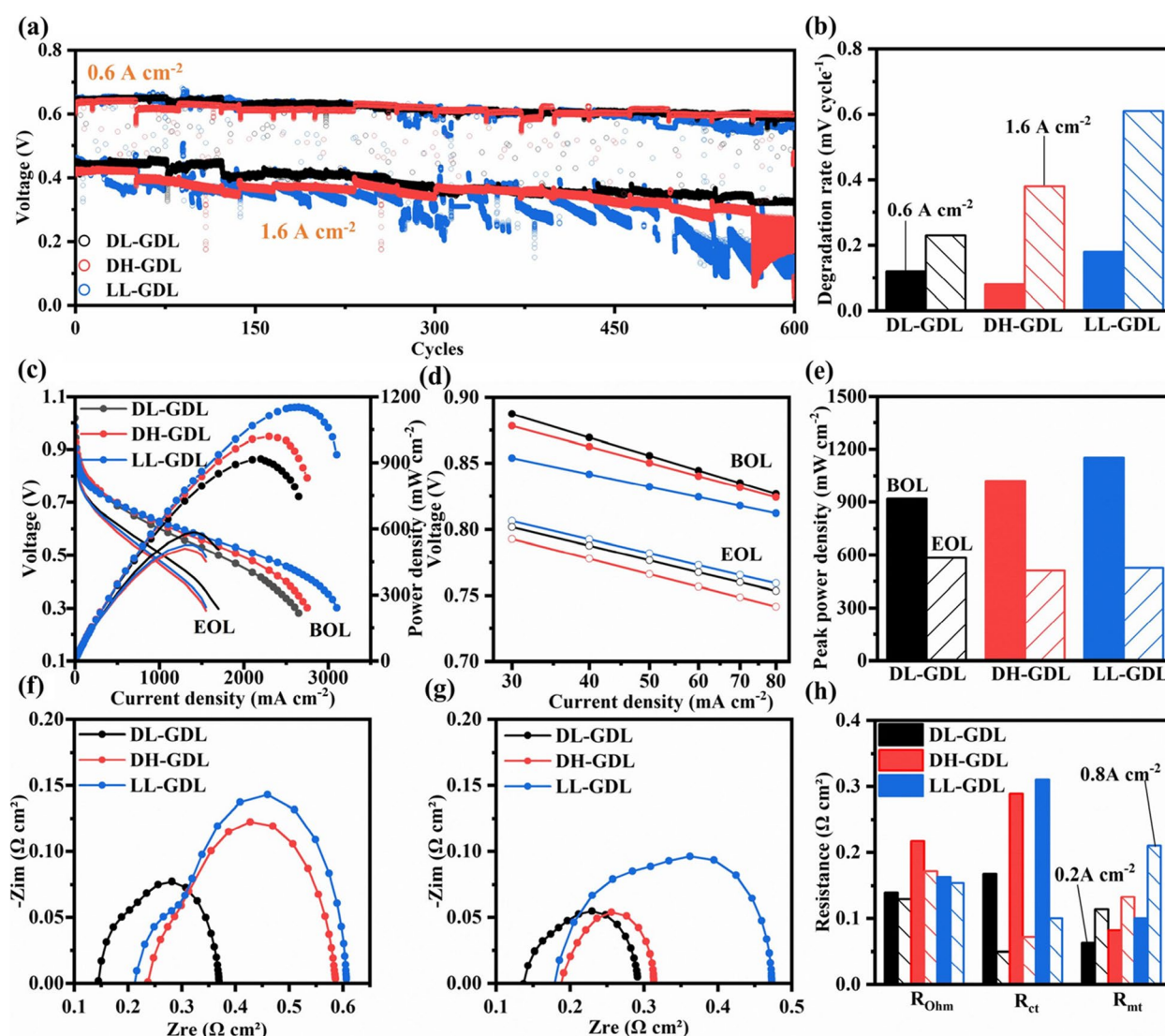
rather than to fully reproduce practical air-breathing duty cycles.

All three single cells operated for more than 200 h (over 600 cycles), as shown in Fig. 4a, and their fuel cell performance is presented in Fig. 4c and Fig. S4. Voltage decay rates at  $1.6\ A\ cm^{-2}$  were  $0.230$  (DL-GDL),  $0.380$  (DH-GDL), and  $0.613$  (LL-GDL)  $mV\ Cycle^{-1}$  (Table S6). Only the DL-GDL cell operated steadily and exhibited a break-in process within 150 cycles, current-induced PA redistribution that homogenized the electrolyte film and enlarged the three-phase boundary. Due to crack-free MPLs, PA escaped only through overlaps between CL cracks and large pores in the MPL. The densest MPL in DL-GDL generated the highest capillary pressure, minimizing liquid PA migrating and preserving the highest electrolyte inventory, resulting in the best stability.

The DH-GDL cell maintained initial operation, showing slow, reversible voltage decline. However, after 564 cycles, irreversible decay set in, and ultra HCD operation could no longer be sustained, implying that a continuous anodic leakage path had formed through CL cracks and large MPL pores. LL-GDL, with the largest MPL pores and the greatest number of crack-pore intersections, could not stabilize PA distribution at  $1.6\ A\ cm^{-2}$ , leading to large voltage fluctuations and eventual performance collapse.

PA lost in the exhaust and adsorbed on bipolar plates (Fig. S13) was quantitatively collected (Table S5). Total loss was lowest for DL-GDL (91 mg), with comparable anodic and cathodic contributions. DH-GDL and LL-GDL exhibited similar total losses (136 and 135 mg, Table S5), but with different anodic/cathodic distributions. Anodic loss dominated for DH-GDL, attributable to its thinner BL, which shortens the leakage path and enhances electromigration, while LL-GDL showed cathodic loss close to DL-GDL, indicating that cathodic vapor loss is governed by current, gas flow rate, and temperature, and is thus almost invariant under the same operating conditions.

Voltage decay during AST was consistent with the performance decline. Post-test peak power densities under  $H_2/O_2$  fell by 36.4% (DL-GDL), 49.8% (DH-GDL), and 54.3% (LL-GDL) (Fig. 4e). Tafel plots (Fig. 4d, Fig. S1, Fig. S5) showed leftward shifts of the intercept. Initial Tafel slopes (Table S4) were  $0.14$ ,  $0.13$ , and  $0.10\ V\ dec^{-1}$  for DL-, DH-, and LL-GDL, respectively; final values converged to  $0.11$ ,  $0.12$ , and  $0.11\ V\ dec^{-1}$ . The decrease in Tafel slope for DL- and DH-GDL may be associated with changes in PA distribution within the electrode, which could influence local reaction environments and reducing local mass transport resistance [36, 37], whereas LL-GDL lost active area and exhibited a slight increase



**Fig. 4** Comprehensive post-AST evaluation of cell durability and degradation mechanisms. **a** Voltage decay profile during AST. **b** Associated degradation rate. **c** BOL/EOL performance comparison. **d** Kinetic analysis via Tafel plots. **e** EOL peak power density (H<sub>2</sub>/O<sub>2</sub>). **f, g** EIS spectra at 200 and 800 mA cm<sup>-2</sup> (H<sub>2</sub>/O<sub>2</sub>, 160 °C). **h** Fitted resistance values

in slope. The observed voltage decay during AST may originate from multiple coupled degradation processes. In addition to PA redistribution and electrolyte depletion, catalyst degradation and structural changes within the electrode may also contribute to the performance decline. However, the clear correlation between PA loss rate and voltage decay suggests that electrolyte migration plays an important role under the present ultra HCD operating conditions.

Electrochemical impedance spectra (EIS) (Fig. 4f–h) aligned with PA loss trends. DL-GDL retained the lowest ohmic resistance due to the highest PA inventory. DH-GDL and LL-GDL, with similar remaining PA,

showed comparable charge-transfer resistance. Mass-transfer resistance differed markedly: despite having the highest initial permeability, the LL-GDL exhibited the largest final diffusion impedance at 0.8 A cm<sup>-2</sup>, which is attributed to PA trapped in abundant MPL-macropore intersection regions and its thicker BL. This trend is also evident from the separated mass-transfer overpotentials (Fig. S2–3 and Fig. S6–7), where the LL-GDL single cell shows a pronounced increase in mass transport overpotential. In contrast, the DL-GDL, with a denser MPL, restricted pore blockage and maintained mass-transfer performance comparable to that of the DH-GDL.

### 3.4 In-operando analysis of PA redistribution

Drawing from the single cell performance and electrochemical analysis, alongside existing research, we have made preliminary speculations regarding the movement of PA within the fuel cell ultra HCD conditions. To further investigate the redistribution of PA during the AST, we developed an in-house in-situ characterization device [38] as shown in Fig. S11 and utilized high-resolution 3D XRM to perform in-situ imaging (Fig. S9-10) of the PA distribution within the MEA under operational conditions (Fig. S12). X-ray computed tomography (CT) scans were conducted in a nondestructive manner, utilizing the varying densities of different materials in the sample to capture images [39]. The experimental setup is shown in Fig S13. The three-dimensional reconstructed images and vertical cross-sectional slices of the MEA are shown in Fig. 5a-c and d-f, respectively (videos shown in the Supporting Information). The membrane, CL, and GDL (carbon fiber portion) can be identified by their specific structures and locations. However, since X-ray absorption is related to atomic number and the absorption levels of phosphorus (P) and carbon (C) are similar [40], we attempted to enhance imaging contrast by adding 10 mol% cesium dihydrogen phosphate ( $\text{CsH}_2\text{PO}_4$ ) [6, 41] during the acid doping process. This improved the imaging of PA distribution in the electrolyte. Nevertheless, much of the PA remains indistinguishable from the GDL. In the images, we treated the GDL/PA as a mixed phase [42], using yellow (cathode) and blue (anode) to represent them. The pores with the least absorption are shown in black, and the metals (Pt and Cs) with the highest absorption are represented in red (cathode) and green (anode). The PBI membrane between the CLs is shown in orange, and small white areas represent the presence of cesium dihydrogen phosphate. Specific volume fraction data is summarized in Table S7, as the current density increases, the volume fraction of the Mix phase at the cathode decreases from 0.13 (initial state) to 0.06 (after HCD operation), while the volume fraction of the Pore phase increases from 0.27 to 0.30. This variation suggests that part of the PA may migrate from the cathode to the anode. However, the volume fraction of the Pore phase at the anode shows only a minor change, suggesting that PA leaching at the anode is partially compensated by PA migration from the cathode, resulting in no significant variation in the overall volume fractions of the Mix phase and the Pore phase. Finally, PA droplets residue was observed in the anode flow channel (Fig. S13) upon disassembly of the experimental setup. It should be noted that the mixed-phase segmentation approach may introduce uncertainty in absolute volume fraction values. Therefore, the

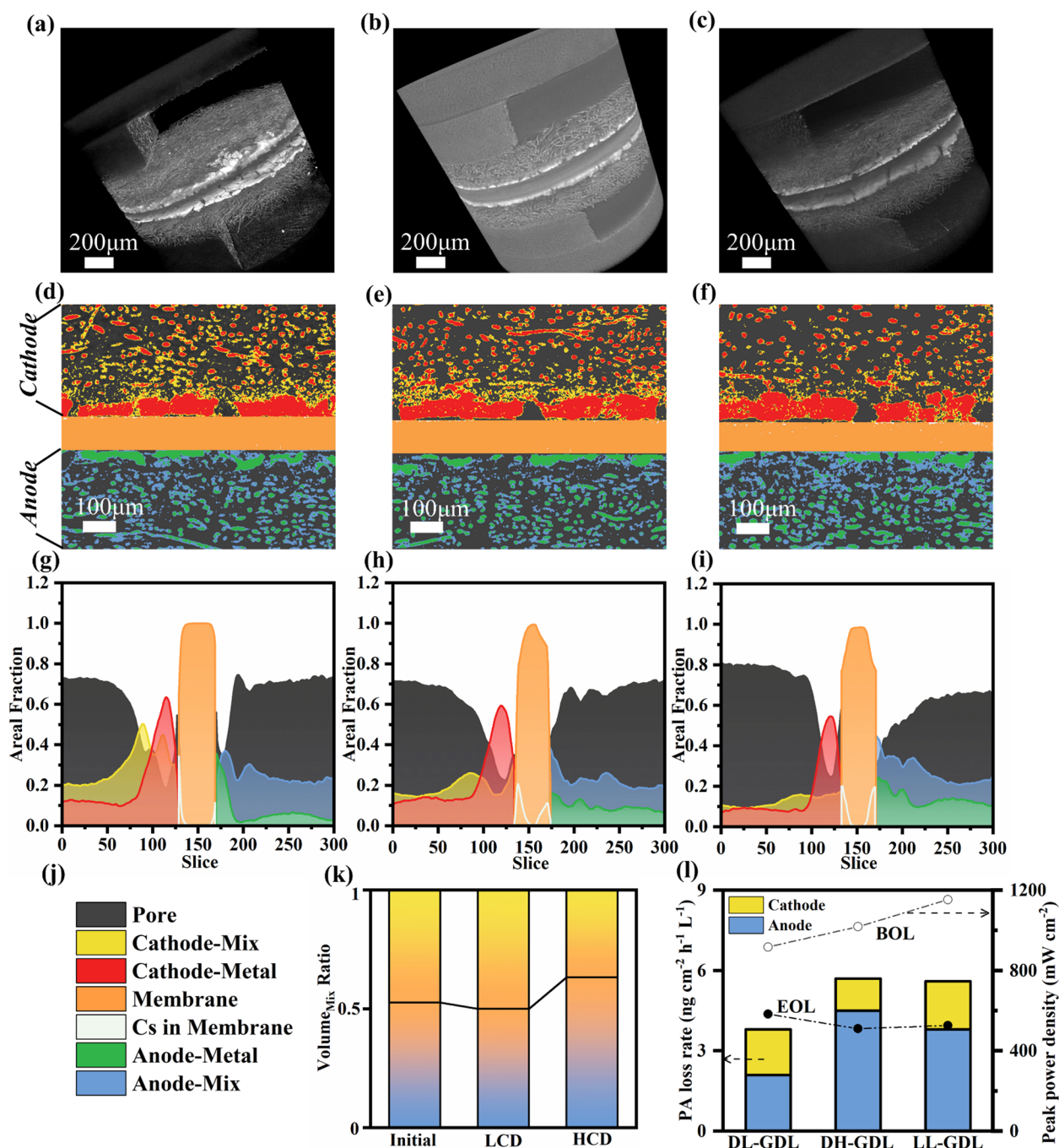
present analysis focuses primarily on the relative evolution of phase distribution during operation rather than exact quantitative values.

From the above results and the PA collection data during the AST, we conclude that the PA migration mechanism varies among different GDL structures under HCD conditions, and is influenced by the microstructure of the MPL, as illustrated in Fig. 6. When the cell operates at HCD for extended periods, electromigration drives PA toward the anode region. A denser MPL reduces the overlap between MPL cracks and CL cracks [43], thereby increasing the capillary barrier for liquid PA penetration and suppressing electrolyte leakage through the GDL. Consequently, the DL-GDL cell exhibits the lowest anodic PA loss rate (Fig. 5I). In contrast, the LL-GDL, which possesses a more loosely structured MPL, provides higher gas permeability but also creates more potential breakthrough pathways for PA migration. In addition to MPL pore structure, variations in BL thickness (Table S1) may also influence the effective leakage pathway. A thinner BL shortens the transport path for PA migration and may partially contribute to the higher anodic acid loss observed for the DH-GDL cell compared with LL-GDL. Although direct visualization of PA migration in all GDL structures remains challenging, the combined electrochemical analysis, acid collection results, and XRM observations consistently indicate that MPL microstructure plays a key role in regulating electrolyte redistribution during operation.

However, increasing MPL density inevitably reduces the gas permeability of the GDL (Table S3). Therefore, an optimized MPL structure should balance the suppression of PA migration with sufficient gas transport capability under HCD operation.

## 4 Conclusions

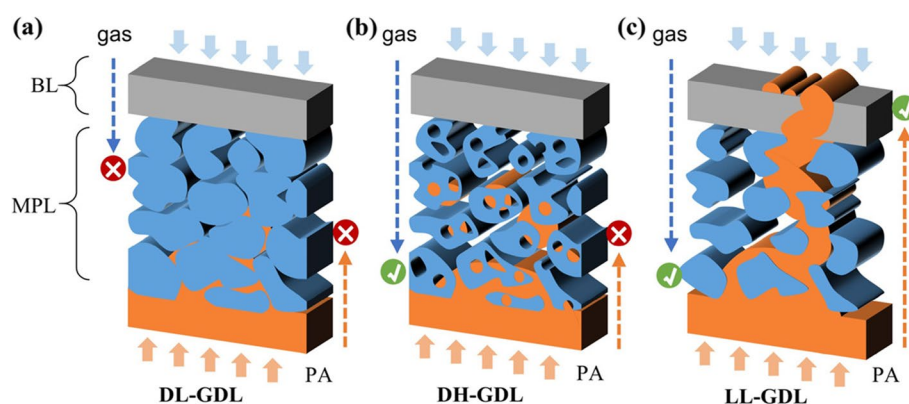
In this work, the influence of GDL porous structure on the long-term stability of HT-PEMFCs under high current density operation is systematically investigated. At the BOL, the fuel cell employing a GDL with a dense MPL delivers balanced electrochemical performance ( $583 \text{ mW cm}^{-2}$ ) compared with other GDL configurations. During AST at  $1.6 \text{ A cm}^{-2}$ , pronounced differences in durability are observed. The GDL featuring a dense and refined MPL exhibits the lowest voltage degradation rate ( $0.23 \text{ mV cycle}^{-1}$ ), whereas cells equipped with less optimized MPL structures suffer from accelerated performance decay, underscoring the critical role of MPL design in sustaining high-current operation. By comparison, the cell employing LL-GDL shows the highest degradation rate ( $0.613 \text{ mV cycle}^{-1}$ ) and a lower PPD of  $526.5 \text{ mW cm}^{-2}$ , corresponding to the poorest stability with a total performance decay of 54.3%. The



**Fig. 5** Comparative analysis of PA redistribution under different operating modes. **a–c** 3D rendered views of PA distribution at BOL **a**, after LCD operation **b**, and after HCD operation **c**. **d–f** 2D segmented cross-sections corresponding to the states in **a–c**. **g–i** Evolution of the localized PA area fraction along the cell thickness for each state. **j** Color legend for **g–i**. **k** Distribution of the mixed-phase (PA/electrode) volume fraction across the tested conditions. **l** Performance-Durability correlation for each cell

DH-GDL presents an intermediate degradation rate of 0.38 mV cycle<sup>-1</sup> and a PPD of 510.9 mW cm<sup>-2</sup>, exhibiting slightly improved stability relative to LL-GDL but still suffering from substantial performance loss (49.8%). These results indicate that a dense MPL with

a refined pore structure can effectively suppress PA loss and maintain the potential for long-term operation under ultra HCD conditions. In-situ XRM analysis further reveals that PA migrates toward the anode and permeates through the GDL, a process strongly



**Fig. 6** Scheme of path of PA Loss in different GDLs. **a** DL-GDL; **b** DH-GDL; **c** LL-GDL

dependent on the applied current loading. The resulting cell performance degradation is closely correlated with the depletion of PA inventory. Overall, this study provides mechanistic insights into electrolyte migration and establishes a practical design strategy for next-generation GDLs aimed at improving the durability and performance of HT-PEMFCs for heavy-duty applications.

### Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43979-026-00169-8>.

Supplementary Material 1.

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### Authors' contributions

Weiwei Ni conceived the study, designed and performed the experiments, conducted formal analysis and data curation, and wrote the original draft. Zhangxun Xia contributed to the conceptualization, methodology, and formal analysis, and reviewed the manuscript. Ruili Sun, Zinan Zhang, Xi Chen, Yangyang Hu, and Jicai Huang contributed to the methodology and conceptual design. Suli Wang acquired funding, participated in conceptualization, and reviewed and edited the manuscript. Gongquan Sun supervised and administered the project, and acquired funding. All authors discussed the results, verified the findings, and commented on the manuscript.

### Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

### Declarations

#### Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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